Ingredients of French face, body and suncare cosmetics

Source: Open Beauty Facts (© Open Beauty Facts contributors, ODbL), French products in the face, body and suncare categories. Downloaded: 15.10.2026.

1. Data preparation

Filtering

- 64,237 products in the dataset; 17,628 with a non-empty ingredient list
- selected country: France: 7,345 (largest amount of records)
  selected categories: face, body and suncare: 1,879 (most suitable for cosmetology review)
  country + selected categories: 1,009
- After list validation: **924 products**, 986 category assignments (62 products carry more than one category tag and are counted in each)

Cleaning of the ingredient text column

- Text length: median 418 characters, 99th percentile 1,359
- Above the 99th percentile: mostly not INCI lists (warnings, instructions, translations); left excluded (<1% of records)
- 95th-99th percentile: mostly just longer, clean INCI lists
- Removed non-ingredient fragments (marketing and legal phrases, addresses, websites, usage instructions)
- 174 records modified; the 6 most heavily trimmed were checked manually, and only extra information was removed
- Lists split into single ingredients, junk removed, name variants unified (dictionary + rules, e.g. aqua/water/eau → water)
- Lists with fewer than 3 ingredients excluded

Classification

- 200 most frequent ingredients classified manually (main class, second class, EU allergen flag)
- Coverage: 68.7% of ingredient occurrences (300 ingredients: 74.8%, 500: 81.2%), so the gain per additional ingredient is small

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2. Dataset

- Face 482 | body 388 | suncare 116 products
- Suncare results are less stable because of the small sample

3. Key findings

Most common ingredients

- Water (741), glycerin (635), fragrance (632)
- Then: tocopherol, linalool, citric acid, limonene, phenoxyethanol, sodium hydroxide, xanthan gum
- 2 of the top 10 are EU fragrance allergens: linalool and limonene
- Same top 3 in every category:
- body: water (321), glycerine (293), fragrance (290)
- face: water (378), fragrance (325), glycerine (301)
- suncare: water (93), glycerine (86), fragrance (64)

Suncare products vs face and body products

- Defined by UV filters (% of products containing the ingredient):

Ingredient	Face	Body	Suncare
butyl methoxydibenzoylmethane	5.0%	1.5%	49.1%
bis-ethylhexyloxyphenol methoxyphenyl triazine	0.2%	0.3%	40.5%
ethylhexyl triazone	1.5%	0.3%	38.8%
octocrylene	2.3%	0.5%	33.6%
ethylhexyl salicylate	1.7%	1.0%	29.3%
drometrizole trisiloxane	0.6%	0.0%	15.5%
homosalate	0.4%	0.0%	14.7%

- Top filters: ~15-50% of suncare products, up to 5% of face, ≤1.5% of body
- Typical supporting ingredients: alcohol denat. (38.8% vs 8.5% face, 3.9% body), C12-15 alkyl benzoate (34.5% vs 1.9%, 1.5%), tocopherol (53.4% vs 29.9%, 35.1%)
- More ingredients per product (avg. 26 vs about 20), consistent with more complex formulas needing to keep UV filters stable

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Face products vs body products

- Very similar composition, differences of a few percentage points
- Body: more fragrance (74.7% vs 67.4%), more caprylyl glycol (20.1% vs 13.5%)
- Face: more alcohol denat. (8.5% vs 3.9%), disodium EDTA (15.1% vs 8.5%), butyl methoxydibenzoylmethane (5.0% vs 1.5%)

Ingredient classes

- Emollients: largest class (42 of 200), then thickeners/stabilisers (26) and emulsifiers (23)
- Highest average reach per ingredient: solvent (145 products), pH buffer (141), fragrance (140); these are means, pulled up by a few very common ingredients (e.g. water)
- Leading ingredients:
    - Active / skin-conditioning: panthenol, aloe barbadensis leaf juice, allantoin
    - Humectants: glycerin, propylene glycol, butylene glycol
    - Emollients: butyrospermum parkii butter, dimethicone, caprylic/capric triglyceride

Ingredients per product

- Average: body 20, face 20.3, suncare 26
- Extremes: body 3-60, face 3-60, suncare 4-51

Fragrance allergens

- Fragrance declared: body 74.7% | face 67.4% | suncare 55.2%
- At least one listed allergen: body 49.2% | face 40.7% | suncare 34.5%
- Lower bound: 14 allergens flagged, only from the top 200 ingredients; benzyl alcohol excluded (mainly used as a preservative)

4. Limitations

- User-contributed data: typos and name variants remain, so counts of some ingredients may be understated
- Classification and allergen flags cover the top 200 ingredients (68.7% of occurrences) only
- INCI lists give no concentrations of each ingredient; a listed ingredient does not mean a harmful product
- Products with several categories are counted in each
- Suncare sample is small (116)
- Colour Index pigments (e.g. CI 77891) were removed by the junk filter, so ingredient counts of tinted products are understated.
- Names ending in a closing bracket lost it during cleaning (255 of 19,219 entries, 138 products), so some variants were not merged (e.g. paraffinum liquidum 106 products plus about 32 in unmerged variants)

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5. Conclusions

- All categories share a core: water, glycerin, fragrance
- Suncare is distinct: UV filters and more complex formulas
- Face and body are very similar
- Fragrance is widespread (55-75% of products), and at least 35-49% of all products declare an EU allergen
- Findings describe the 200 most common ingredients, which is enough for rankings and category comparisons

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